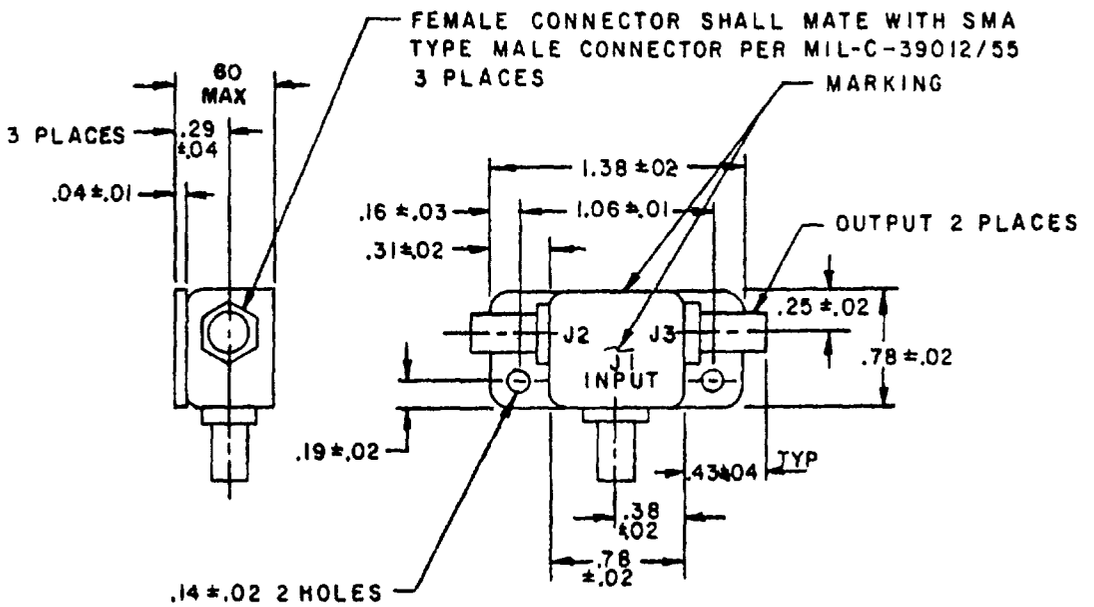
MILITARY SPECIFICATION SHEET

POWER DIVIDER/COMBINERS, N-WAY, 0 DEGREES, SMA CONNECTORS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the power divider/combiners described herein shall consist of this document and the latest issue of specification MIL-P-23971.



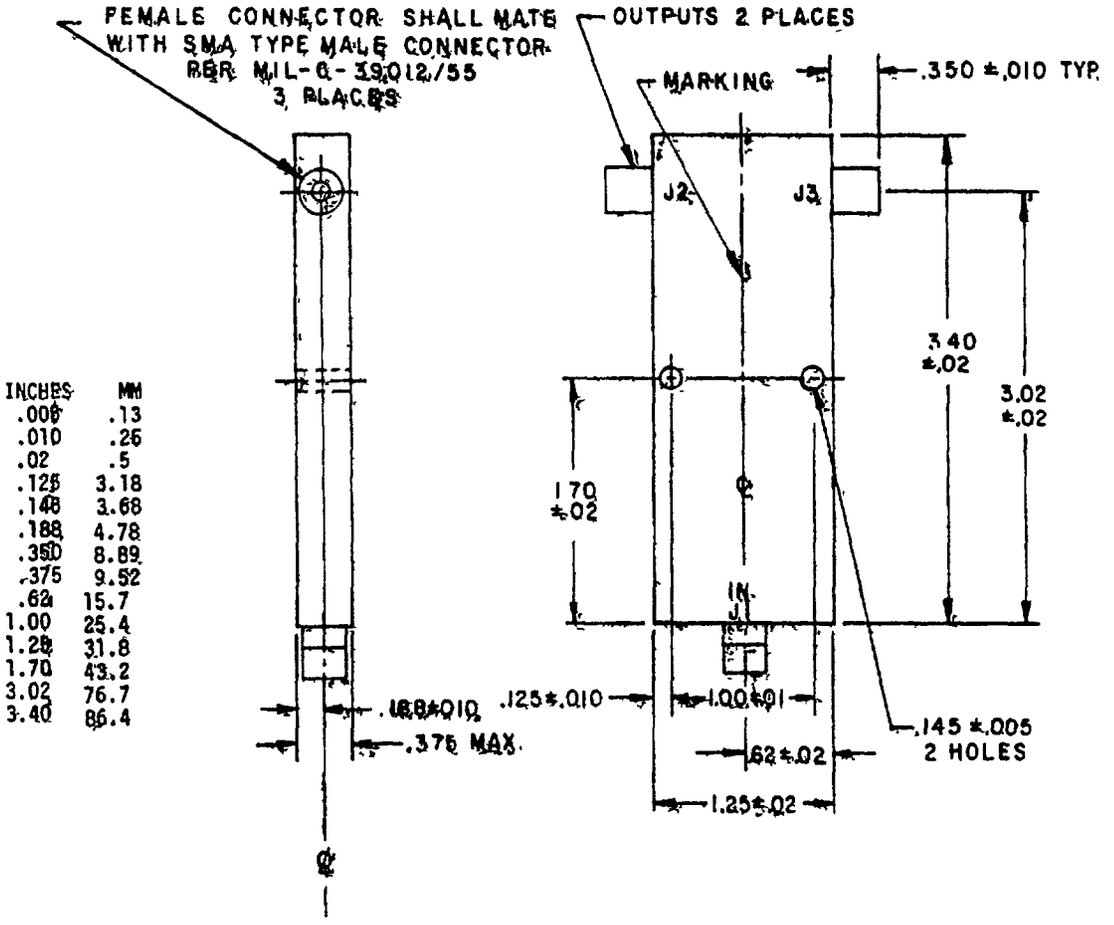
INCHES	MM	INCHES	MM
.01	.3	.29	7.4
.02	.5	.31	7.9
.03	.8	.38	9.7
.04	1.0	.43	10.9
.14	3.6	.50	12.7
.16	4.1	.78	19.8
.19	4.8	1.06	26.9
.25	6.4	1.38	35.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

FIGURE 1. Dimensions and configuration, 2-way, dash number 01.

FSC 5985

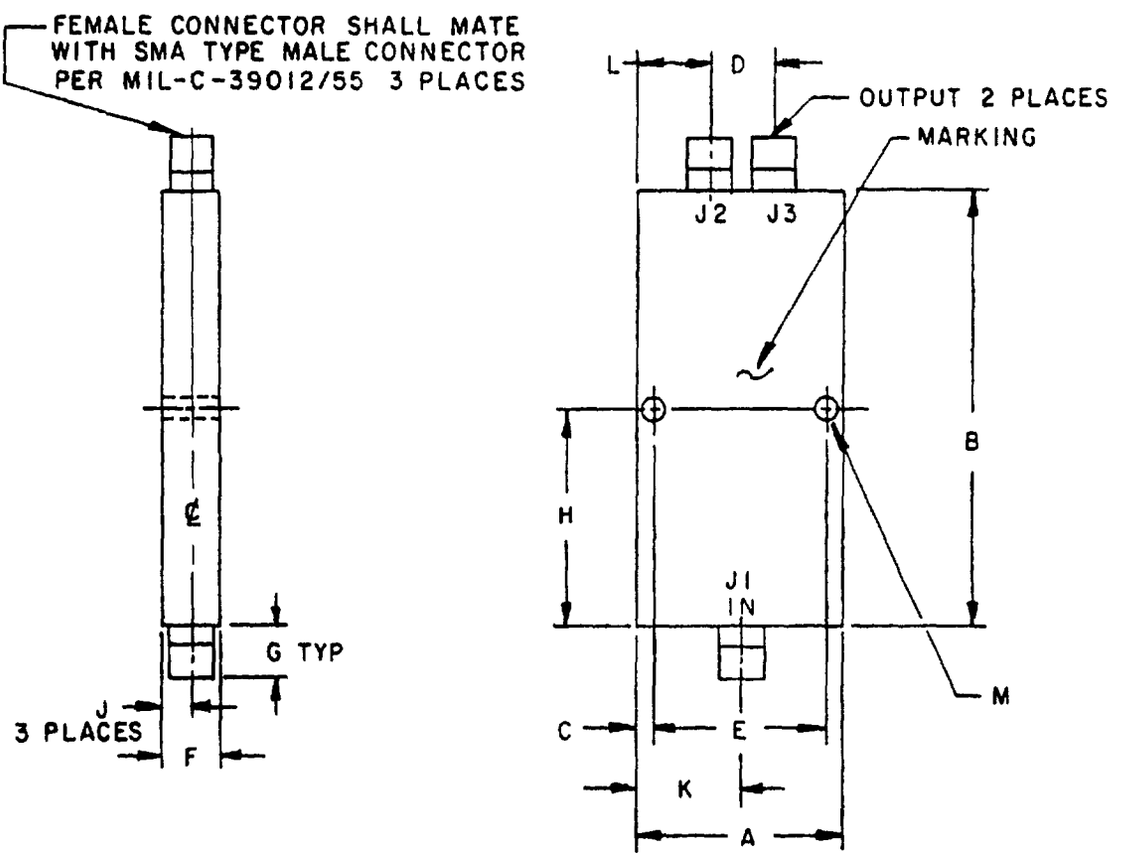
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

FIGURE 2. Dimensions and configuration, 2-way, dash number 02.

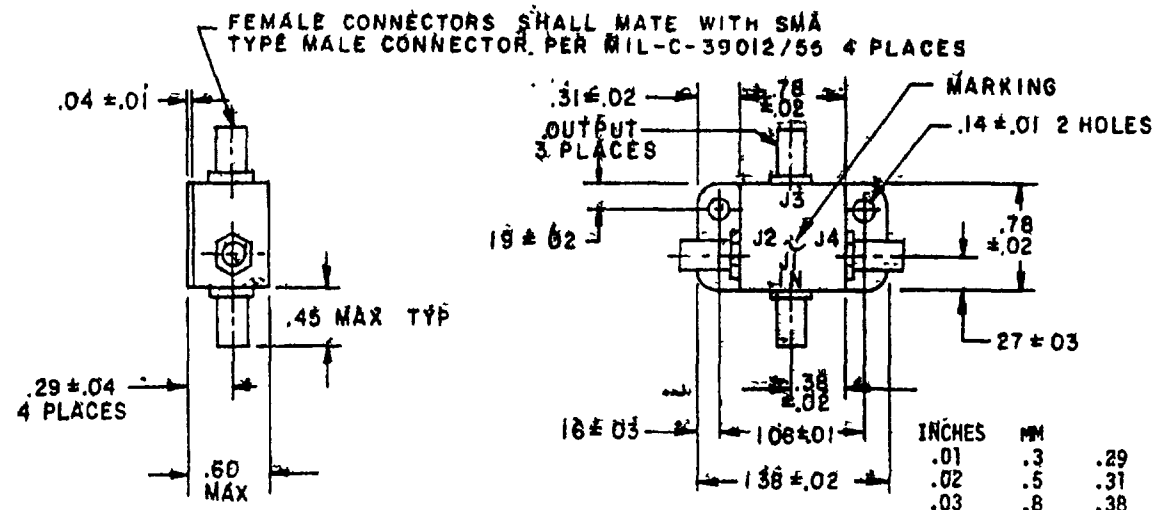
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 3. Dimensions and configuration, 2-way dash numbers 03 and 04.

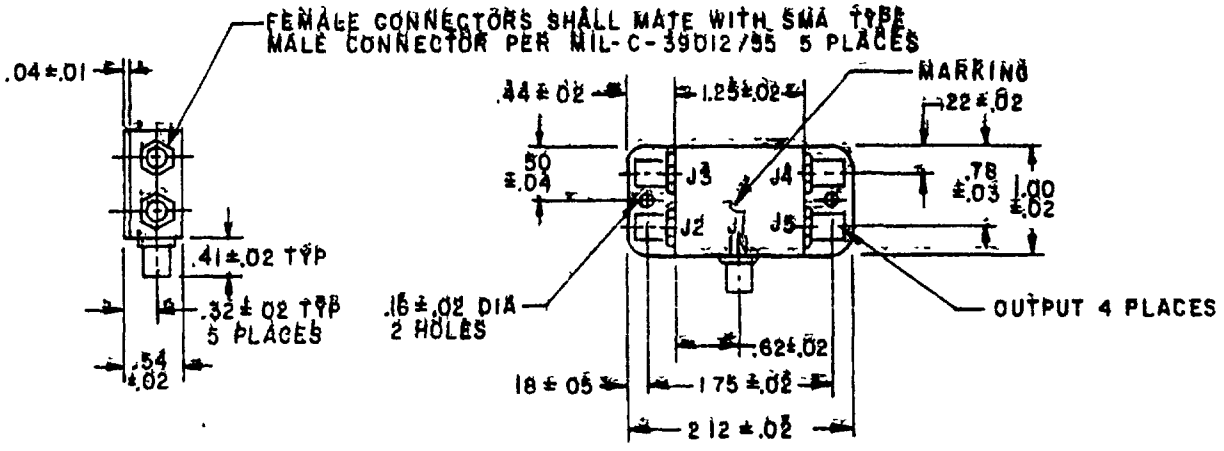
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INCHES	MM		
.01	.3	.29	7.4
.02	.5	.31	7.9
.03	.8	.38	9.7
.04	1.0	.45	11.4
.14	3.6	.60	15.2
.16	4.1	.78	19.8
.19	4.8	1.06	26.9
.27	6.9	1.38	35.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 4. Dimensions and configuration, 3-way, dash number 05.

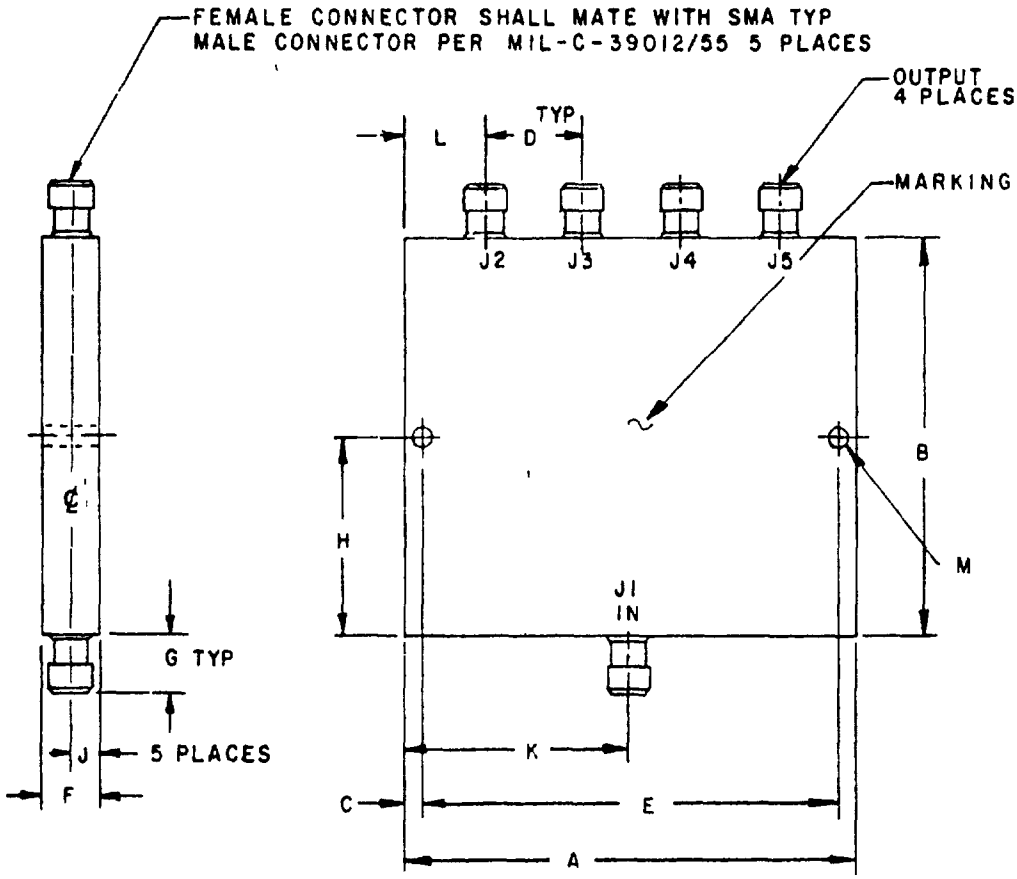


INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.01	.3	.16	4.1	.44	11.2	1.00	25.4
.02	.5	.18	4.6	.50	12.7	1.25	31.8
.03	.8	.22	5.6	.54	13.7	1.75	44.4
.04	1.0	.32	8.1	.62	15.7	2.12	53.8
.05	1.3	.41	10.4	.78	19.8		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 5. Dimension and configuration, 4-way, dash numbers 06, 07, 08, 09, 10, and 11.

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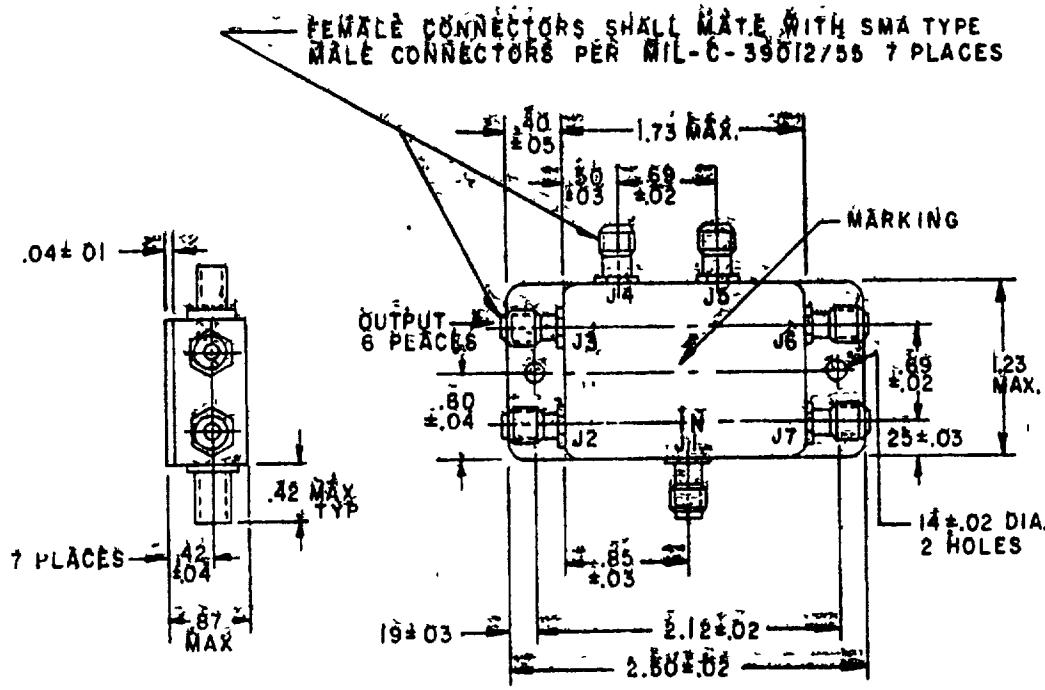


INCHES	MM	INCHES	MM
.005	.13	.55	14.0
.01	.3	.60	15.2
.02	.5	.70	17.8
.03	.8	.75	19.0
.125	3.18	1.20	30.5
.142	3.61	1.40	35.6
.145	3.68	1.60	40.6
.18	4.6	2.40	61.0
.35	8.9	2.55	64.8
.36	9.1	2.80	71.1
.37	9.4	2.85	72.4
.45	11.4	2.95	74.9
.50	12.7	3.20	81.3

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 6. Dimensions and configurations , 4-way, dash numbers 12, 13, and 14.

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INCHES	MM	INCHES	MM
.01	.3	.50	12.7
.02	.5	.60	15.2
.03	.8	.69	17.5
.04	1.0	.85	21.6
.05	1.3	.87	22.1
.14	3.6	1.23	31.2
.19	4.8	1.73	43.9
.25	6.4	2.12	53.8
.40	10.2	2.50	63.5
.42	10.7		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 7. Dimensions and configuration, 6-way, dash number 15.

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Table I Dimensions

M23971/16-	A	B	C	D	E	F	G	H	J	K	L	M	Figure
03,04	1.45 ±.02	3.00 ±.02	.120 ±.030	.75 ±.02	1.20 ±.01	.37 ±.02	.35 ±.03	1.500 ±.020	.185 ±.010	.72 ±.02	.35 ±.02	.145 ±.005	3
12	3.20 ±.02	2.85 ±.02	.125 ±.030	.70 ±.02	2.95 ±.01	↓	↓	.142 ±.020	↓	1.60 ±.02	.55 ±.02	↓	6
13	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	6
14	2.80 ±.02	2.40 ±.02	↓	.60 ±.02	2.55 ± .01	.36 ±.02	↓	1.200 ±.020	.180 ±.010	1.40 ±.02	.50 ±.02	↓	6

INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.35	8.89	1.40	35.6
..01	.3	.36	9.14	1.45	36.8
.02	.5	.37	9.40	1.500	38.10
.020	.51	.50	12.7	1.60	40.6
.030	.76	.55	14.0	2.40	61.0
.120	3.05	.60	15.2	2.55	64.8
.125	3.17	.70	17.8	2.80	71.1
.142	3.61	.72	18.3	2.85	72.4
.145	3.68	.75	19.1	2.95	74.9
.180	4.57	1.20	30.5	3.00	76.2
.185	4.70	1.200	30.48	3.20	81.3

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REQUIREMENTS:

Design and construction: See figures 1 through 7.

Case: O-ring seal.

Electrical characteristics: See table II.

Weight: See table II.

Ambient temperature: See table II.

Environmental requirements: See table III.

Part number: M23971/16– (dash number from table II).

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TABLE II Electrical performance characteristics and physical requirements.

M23971/16	Impedance (ohms)	Frequency range GHz	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance max (degree)	Amplitude balance (dB) max	Power level		Weight pounds (grams)	Ambient temperature		Figure
									avg (W)	pk (W)		Operating	Storage	
01	50	1-100 MHz	3.01 +.5 −.0	1.3:1	0 5	30	±1 0	±0.2	–	2	.062 (28)	−55° to +85°C	−65° to +85°C	1
02	50	1 0-2.0	3 + 2 −.0	1.25.1	0.3	20	±6 0	±0 4	–	.1	.17 (77 0)	0° to +71°C	0° to +71°C	2
03	50	1.0-2.0	3 +.2 −.0	1.25·1	0.3	20	±6 0	±0 4	–	.1	.17 (77.0)	0° to +71°C	0° to +71°C	3
04	50	1.0-4.0	3 +.2 − 0	1.7:1	0 40	22	±4.0	±0 2	1	1/	.25 (110.)	−35° to +105°C	−55° to +125°C	3
		4.0-8 0		1 6 1	0 80									
		8 0-12.0			1.20	16	±5 0							
		12.0-12.4			1.30									
05	50	10-100 MHz	4.77 + .5 −.0	1.3 1	0 5	30	±1 0	±0 2	–	5	.120 (55)	−55° to +85°C	−65° to +85°C	4
06	50	2-32 MHz	6. +.5 −.0	1.3·1	0 7	30	±1.0	±0.2	–	2	06 (27 0)	−55° to +72°C	−62° to +110°C	5
07	50	10-100 MHz	6 02 + 7 − 0	1.3 1	0 7	30	±1.0	±0 2	–	2	.06 (27.0)	−55° to +72°C	−62° to +110°C	5
08	50	20-200 MHz	6 +.5 − 0	1 3 1	0 7	30	±1 0	±0.2	–	2	.06 (27.0)	−55° to +72°C	−62° to +110°C	5
09	50	100-300 MHz	6 + 5 −.0	1 3.1	1 0	30	±1 0	±0 2	–	2	06 (27 0)	−55° to +72°C	−62° to +110°C	5
10	50	200-500 MHz	6. + 5 − 0	1 3 1	1 0	30	±1 0	±0 2	–	2	06 (27 0)	−55° to +72°C	−62° to +110°C	5

See notes at end of table

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TABLE III. Electrical performance characteristics and physical requirements – Continued.

M23971/16 –	Impedance (ohms)	Frequency range GHz	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance max (degree)	Amplitude balance (dB) max	Power level avg (W)	Power level pk (W)	Weight pounds (grams)	Ambient temperature Operating	Ambient temperature Storage	Figure
11	50	250-500 MHz	6 + .5 − 0	1.3:1	1.0	30	±1.0	±0.2	–	2	.062 (28)	−54° to +71°C	−57° to +71°C	5
12	50	1-2	6 + .5 − 0	1.35:1	0.6	20	±6.0	±0.6	–	1	.37 (170)	0° to +71°C	0° to +71°C	6
13	50	1-2	6 + .9 − 0	1.35:1	0.6	20	±6.0	±0.6	3 1/ 16 2/		.36 (160)	0° to +71°C	0° to +71°C	6
14	50	2-4	6 + .5 − 0	1.35:1	0.6	18	±3.0	±0.3	.1	–	.360 (162)	−54° to +71°C	−57° to +71°C	6
15	50	10-100 MHz	7.8 + .7 − 0	1.3:1	1.0	30	±2.0	±0.2	2	–	.19 (160)	−46° to +63°C	−57° to +7°C	7

1/ Power level: 275 watt, peak, minimum for 100 nanoseconds.
2/ Power level: When used as a power combiner, the maximum power to each port is .079 watts for coherent (equal amplitude, phase and frequency) signals. For 180° out of phase signals power rating is reduced to .02 watts per port.

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TABLE III Environmental tests: In accordance with MIL-P-23971 except:

M23971/16-	Vibration	Shock	Salt spray	Explosion	Acceleration
01, 05, 06 07, 08, 09 10	test condition B	4/	test condition A	4/	4/
04, 15	4/	test condition A	4/	4/	4/
11	4/	4/	4/	applicable	3/
13	4/	test condition I	test condition A	4/	4/
14	4/	test condition I	4/	4/	4/

3/ The item shall be tested in accordance with method 212 of MIL-STD-202, test condition A with a 6G level. The acceleration measured at any point of the component shall not exceed a 0.9G level. After the test, the VSWR, insertion loss, isolation, phase balance, and amplitude balance tests shall be performed in accordance with 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8, respectively.
4/ Not applicable.

TABLE IV Supersession data

Part number M23971/16-	Supersedes M15370/13-
01	001
05	002
07	003

NOTE. Power dividers covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash Number	Manufacturer's code	Manufacturer's part number	FSN
01	99971 12457	7541409-1 PDM-20-50	
02	05869 31597	712439-1 40265	
03	05869 31597	712439-2 4J0265	
04	12436 31597	6010207-001 40510	

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Note – continued.

Dash Number	Manufacturer's code	Manufacturer's part number	FSN
05	99971 12457	7541409-2 PDM-30-55	
06	05869 12457	721723-1 PDM-40-17/12030	
07	99971 12457 05869 12457	7541409-3 PDM-40-55 721723-2 PDM-40-55/12030	
08	05869 12457 04660 12457	721723-3 PDM-40-110/12030 A600281-001 PDM-40-110	
09	05869 12457	721723-4 PDM-40-200/12030	
10	05869 12457	721723-5 PDM-40-350/12030	
11	32562 12457	340-1002-010 PDM-40-375	
12	05869 31597	712443-1 40275	
13	10001 31597	2939101-1 4A0275	
14	32562 31597	340-1001-010 40276	
15	49956 12457 32252	934530-1 PDM-60-55/23047 0-HJ-6-7010	

Custodians:
Army – ER
Navy – EC
Air Force – 85

Review activities:
Army –
Navy –
Air Force –
DLA – ES

User activities:
Army –
Navy – AS, OS, MC
Air Force –

Preparing activity:
Navy – EC

Agent:
DLA – ES

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DOCUMENT IDENTIFIER AND TITLE Mil-P-23971/16

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CONTRACT NUMBER

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2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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